

Task 6: K-Nearest Neighbors (KNN) Classification

★ Task 6: K-Nearest Neighbors (KNN) Classification – Full Explanation

✓ Objective

To understand and implement the **K-Nearest Neighbors (KNN)** algorithm for solving **classification problems** using Python and machine learning libraries.

KNN is a **supervised learning algorithm** that classifies new data based on the **majority class of its nearest neighbors**.

🔧 Tools Required

You will use these Python libraries:

1. Scikit-learn (sklearn)

For loading datasets

For the KNeighborsClassifier

For accuracy & confusion matrix metrics

For train-test split

2. Pandas

For loading and handling datasets (CSV, tables)

3. Matplotlib

For plotting graphs

For visualizing decision boundaries

Steps / Mini Guide (Explained)

1. Choose a classification dataset and normalize features

You can use:

Iris dataset

Wine dataset

Breast cancer dataset

Or your own CSV file

Normalization is needed because KNN uses **distance**, and distance is affected by scale.

Example:

```
from sklearn.preprocessing
import StandardScaler
```

2. Use KNeighborsClassifier from sklearn

Import the model:

```
from sklearn.neighbors import
KNeighborsClassifier
```

Create KNN model:

```
knn =
KNeighborsClassifier(n_neighbors=5)
```

3. Experiment with different K values

Try K = 3, 5, 7, 9, etc.

This helps you find the **best performance**.

4. Evaluate your model

Use:

Accuracy

Confusion Matrix

Classification Report

Example:

```
from sklearn.metrics import
accuracy_score,
confusion_matrix
```

5. Visualize decision boundaries

Only possible for **2 features** (2D plot).
You can plot how the model separates different classes.

What You Will Learn from This Task

- ✓ How KNN works (distance-based classification)
- ✓ How feature scaling affects performance
- ✓ How to choose the right value of K
- ✓ How to evaluate classification models
- ✓ How to visualize ML decision regions

Simple Example Output You Will Produce

Train KNN model

Print accuracy

Show confusion matrix

Plot decision boundary

If you want, I can also provide:

- ✓ Full Python code for this task
- ✓ Step-by-step explanation
- ✓ Dataset recommendation
- ✓ Graph + confusion matrix code

Just tell me!